F2 AI 工作坊 3 / Workshop 3

CNN 與圖像識別
CNN and image recognition

用圖片教 iPad 分辨文具

Teach an iPad to recognise stationery

鉛筆、原子筆、擦膠；不需編程

Pencil, pen and eraser; no coding

今天的挑戰

Today’s challenge

認識圖像分類，試玩 CNN 示範

Explore image classification and try a CNN demo

用其他組別的文具訓練模型

Train a model using other groups’ stationery

投影 iPad，測試自己組別的全新文具

Mirror the iPad and test your group’s unseen items

回歸與分類

Regression and classification

回歸：預測數值，例如車程是 18.5 分鐘

Regression: predict a value, such as a journey time of 18.5 minutes

分類：選擇類別，例如鉛筆、原子筆或擦膠

Classification: choose a category, such as pencil, pen or eraser

手寫數字「7」也可以是類別標籤

A handwritten digit “7” can also be a category label

快問快答：數值還是類別？

Quick vote: value or category?

A：預測明天的氣溫

A: predict tomorrow’s temperature

B：辨認鉛筆、原子筆或擦膠

B: recognise a pencil, pen or eraser

C：辨認圖片中的數字 0 至 9

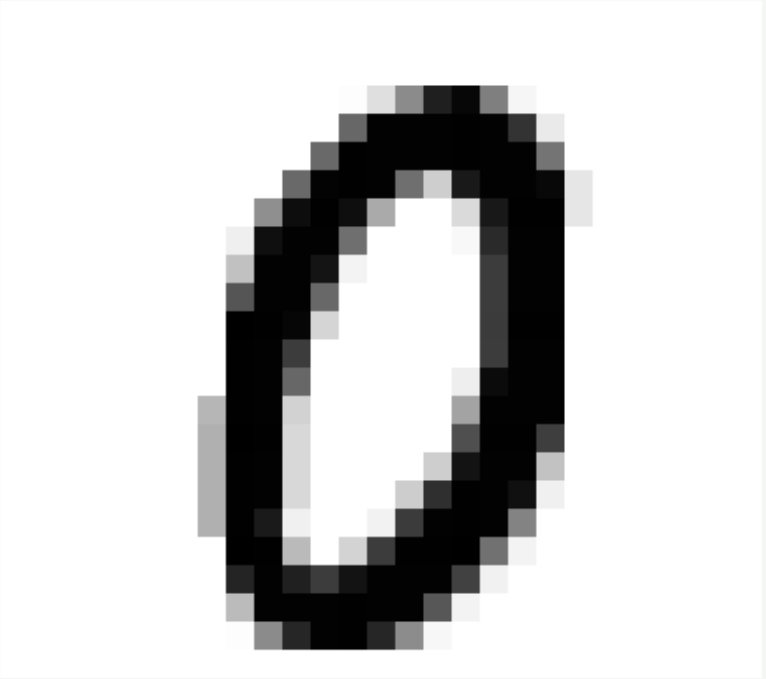
C: identify which digit, 0–9, appears in an image

圖片是一格格像素

An image is a grid of pixels

我們看到數字；電腦收到明暗數值

We see a digit; the computer receives brightness values



模型從有標籤的例子學習模式

The model learns patterns from labelled examples

CNN 怎樣辨認圖片？

How does a CNN recognise images?

濾鏡掃描小區域，找邊緣等局部模式

Filters scan small patches for patterns such as edges

把局部模式組合成更複雜的特徵

Combine local patterns into more complex features

用這些特徵，為可能的類別評分

Use these features to score possible classes

CNN：卷積神經網絡；濾鏡是學習出來的模式偵測器

CNN: convolutional neural network; a filter is a learned pattern detector

CNN 為甚麼重要？

Why did CNNs matter?

同一濾鏡可在圖片不同位置尋找模式

The same filter can find patterns in different positions

從例子學習特徵，減少逐一手寫辨認規則

Learn features from examples instead of hand-coding every rule

更多數據及更快硬件，幫助辨認複雜照片

More data and faster hardware helped recognise complex photos

CNN 的幾個重要里程碑

A few CNN milestones

1989：辨認手寫郵政編碼

1989: early practical recognition of handwritten postcodes

1998：LeNet-5，數字識別的重要網絡

1998: LeNet-5, a landmark network for digit recognition

2012：AlexNet 勝出 ImageNet 比賽

2012: AlexNet wins ImageNet, a major leap on real-world photos

MNIST：手寫數字挑戰

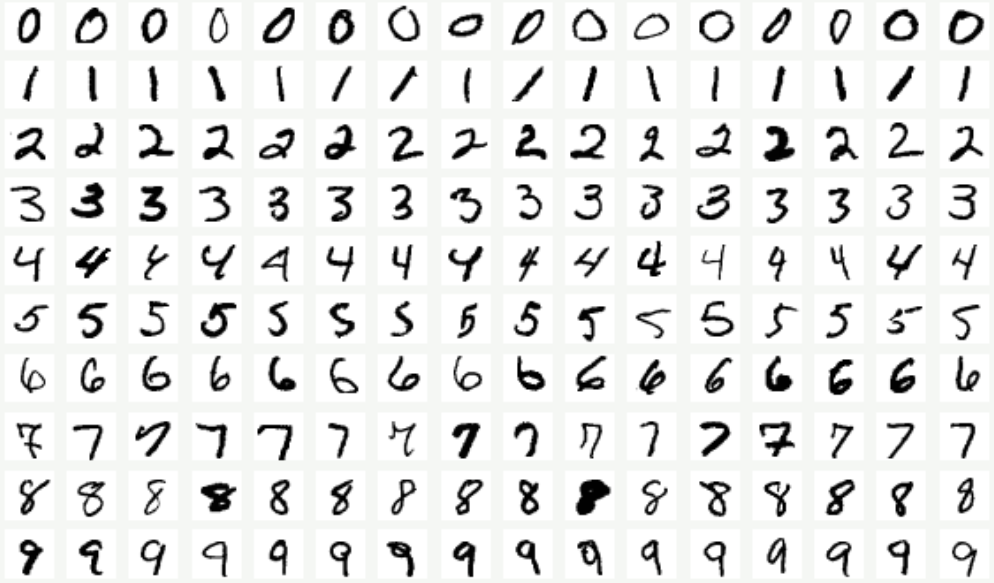
MNIST: a handwriting challenge

10 個類別：0–9；每張是 28 × 28 像素

10 classes, 0–9; each image is 28 × 28 pixels

60,000 張訓練圖片；10,000 張測試圖片

60,000 training images; 10,000 separate test images

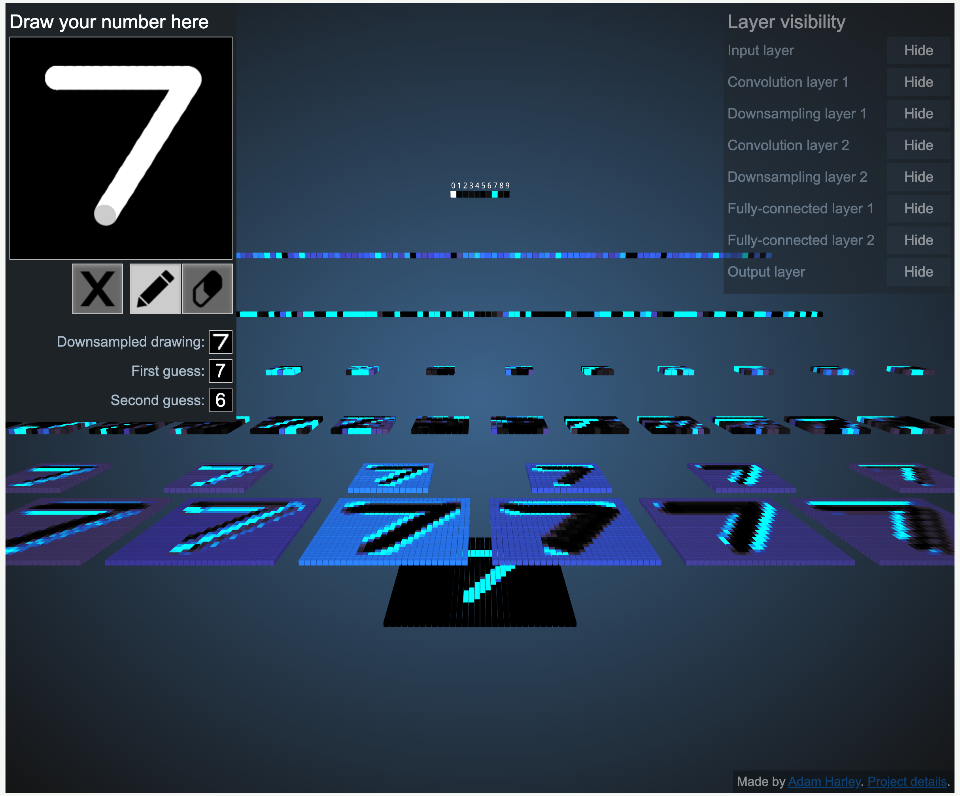


試玩 CNN 網上示範

Try the live CNN demo

畫一個大數字，看 First guess 和 Second guess

Draw a large digit. Compare the first and second guesses



這個模型已訓練好；重畫只是改變輸入

This model is already trained; redrawing changes its input

adamharley.com/nn_vis/cnn/3d.html

重畫後，甚麼改變了？

What changes when you redraw?

先畫工整的 7，再畫傾斜的 7

Draw a neat 7, then a tilted 7

比較兩個猜測，觀察亮起的網絡區域

Compare the guesses and watch which regions light up

模型總是答對嗎？甚麼可能令它認錯？

Is it always right? What might make it struggle?

訓練與測試，工作不同

Training and testing have different jobs

訓練：用例子和正確標籤，學習模式

Training: learn patterns from examples and correct labels

測試：用未參與訓練的新例子，檢查結果

Testing: check results using new examples kept out of training

如同做練習後，再做一題未見過的題目

Like practising, then trying a question you have not seen

用 Teachable Machine 教電腦分類

Teach a classifier with Teachable Machine

Google 的網頁工具，不需編程

A web tool from Google; no coding needed

借用圖像模型，再教它我們的三個類別

Reuse an image model and teach it our three classes

收集圖片 → 訓練 → 預覽及測試

Gather images → train → preview and test

步驟 1：準備小組

Step 1: set up your group

把鉛筆、原子筆、擦膠放在已標示組別的桌上

Place a pencil, pen and eraser on your labelled desk

分工：iPad 操作、拍照、數據檢查

Choose an iPad operator, photographer and data checker

留給其他組別拍照；自己組別不要用來訓練

Other groups may photograph them; your group must not train on them

最重要的規則：保留公平測試

The key rule: keep the test fair

其他組別的文具：作為我們的訓練例子

OTHER groups’ items: use these for our training

自己組別的文具：留待模型訓練完成後測試

OUR group’s items: keep these unseen until testing

不同角度，仍然是同一件文具；不要混入訓練

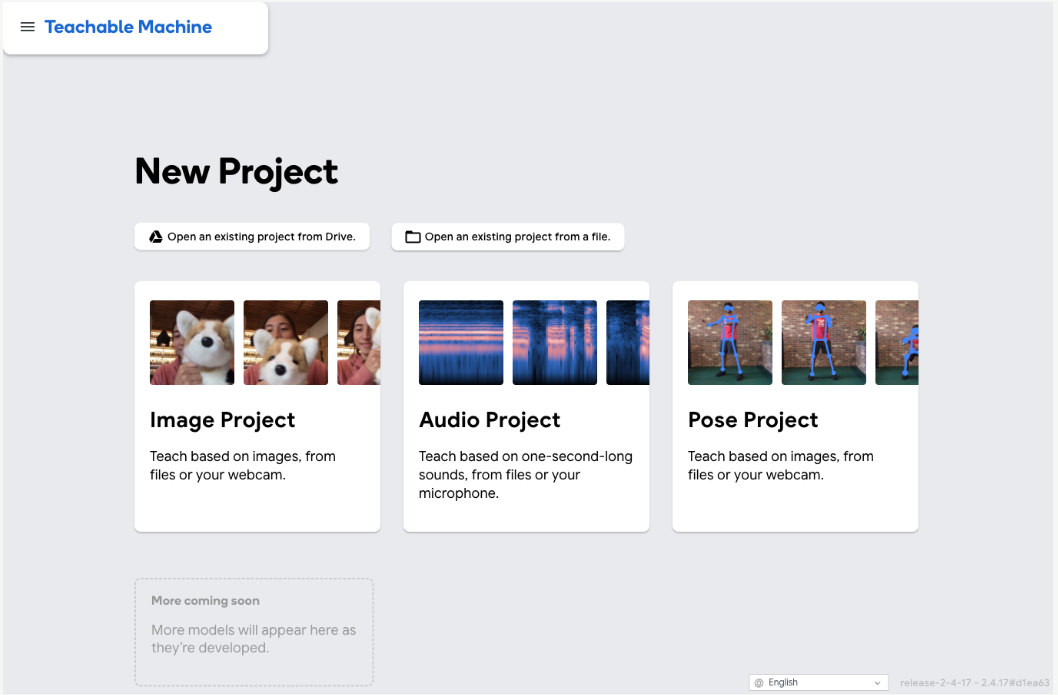
A new angle is still the same item; keep our items out of training

步驟 2：開啟 Teachable Machine

Step 2: open Teachable Machine

iPad 橫向，使用 Safari 開啟下方連結

Use Safari on the iPad in landscape



按 Image Project（圖像專案）

Tap Image Project

teachablemachine.withgoogle.com/train

步驟 3：選擇圖像模型

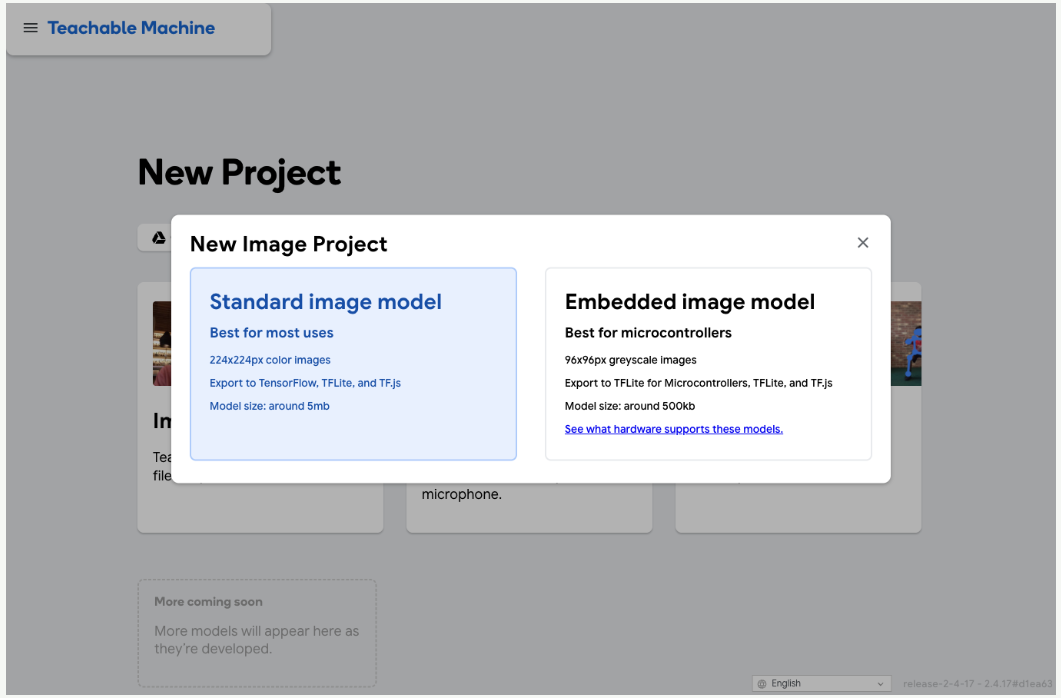
Step 3: choose the image model

按 Standard image model

Tap Standard image model

選標示 Best for most uses 的選項

Choose the option marked “Best for most uses”



步驟 4：設定三個類別

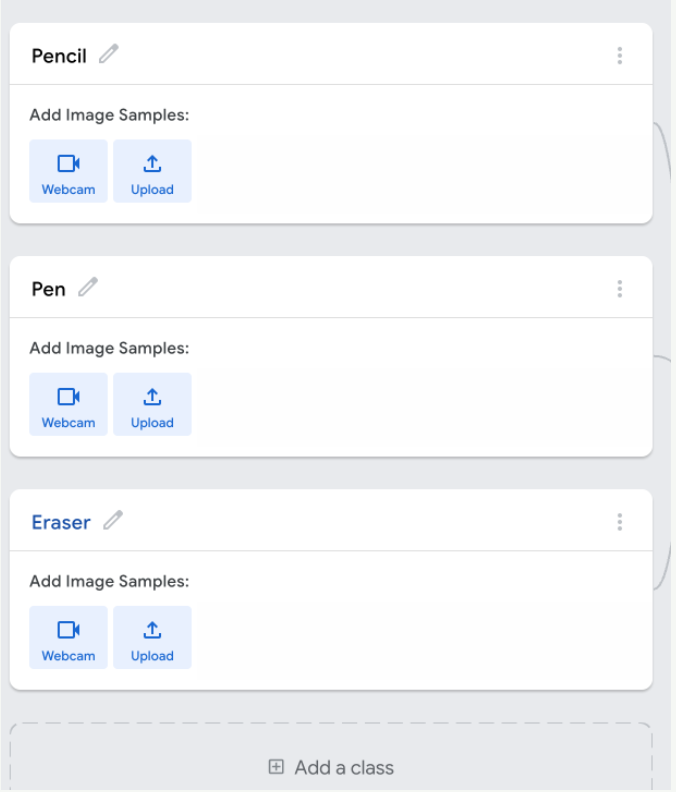
Step 4: name three classes

按名稱旁的筆形圖示，設定 Pencil、Pen

Use the edit icon to name Pencil and Pen

按 Add a class，第三類命名 Eraser

Tap Add a class and name the third class Eraser



步驟 5：拍攝其他組別的文具

Step 5: photograph other groups’ items

用相機，到至少三個其他組別的桌子拍攝

Use Camera at three or more OTHER groups’ desks

目標每類約 30 張，三類數目相近

Aim for about 30 photos per class, with similar counts

每次一件物件；不要拍入自己的文具

One item at a time; keep your own stationery out of the frame

同一物件，要拍攝多個角度

Photograph the same object from several angles

上方、正面、側面，再轉動物件

Try above, front, side and rotated views

移動 iPad，稍微改變距離及光線

Move the iPad; vary distance and lighting a little

整件物件置中；也要拍攝多件不同文具

Keep the whole item centred; include several different items too

讓模型學習文具本身

Help the model learn the object

圖片清晰，物件完整，放在中央

Use clear pictures with the whole object in the centre

每類都使用相近的背景變化

Use similar background variety for every class

刪去模糊、錯類別或拍入自己文具的圖片

Delete blurry, wrongly labelled or accidental own-item photos

步驟 6：上載到正確類別

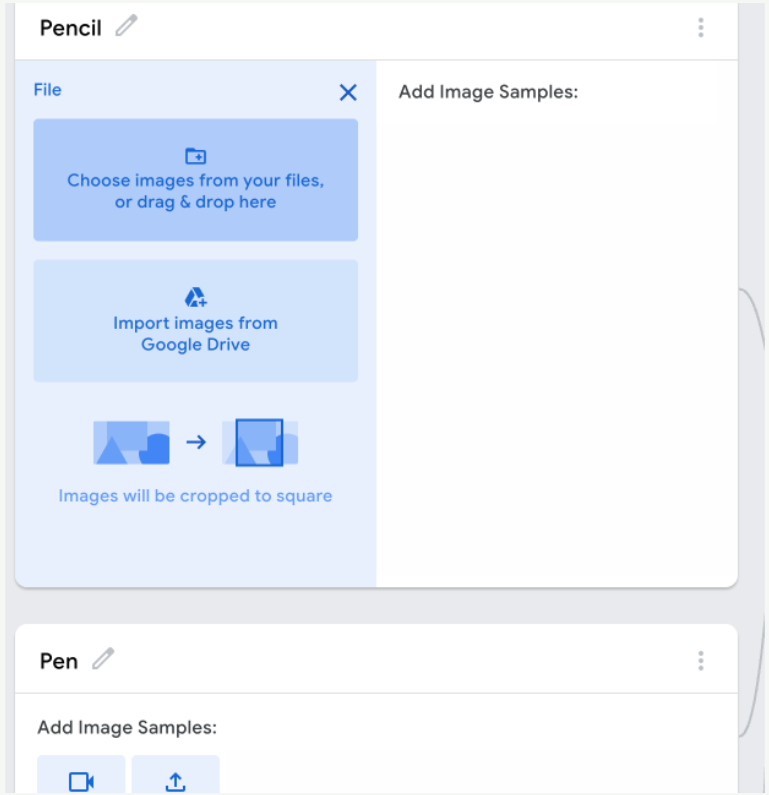
Step 6: upload into the correct class

在 Pencil 按 Upload，再按 Choose images…

In Pencil, tap Upload, then “Choose images from your files”

從相片圖庫或檔案選鉛筆照片，再重複另兩類

Choose pencil photos from Photos or Files; repeat for the other classes



步驟 7：檢查訓練圖片

Step 7: check the training data

標籤正確？三類的數目相近？

Correct labels? Similar counts in all three classes?

多件物件？同一物件有多個角度？

Several objects? Several angles of the same object?

自己組別的測試文具，完全沒有混入？

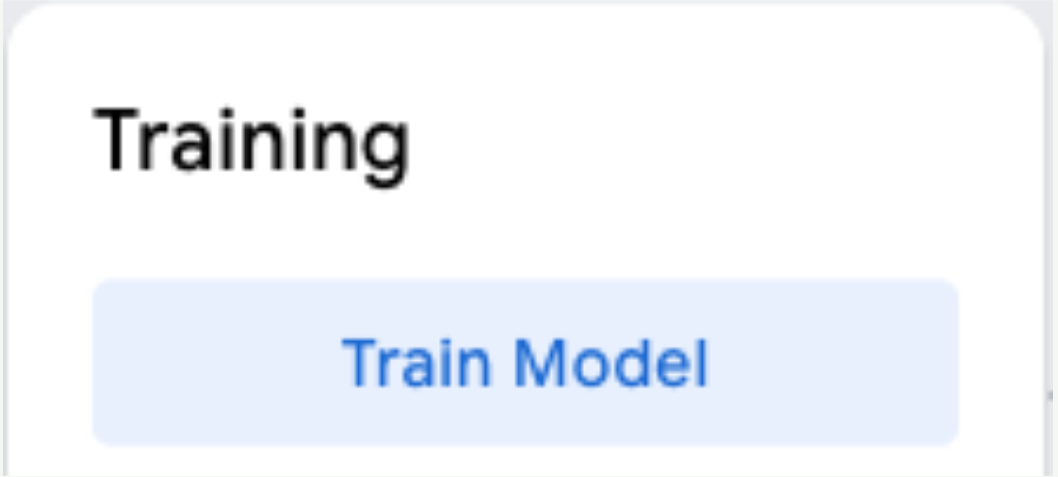
Zero images of our group’s own test stationery?

步驟 8：訓練模型

Step 8: train the model

按 Train Model；保留預設設定

Tap Train Model; leave the default settings



等待完成；保持頁面開啟，不切換或鎖屏

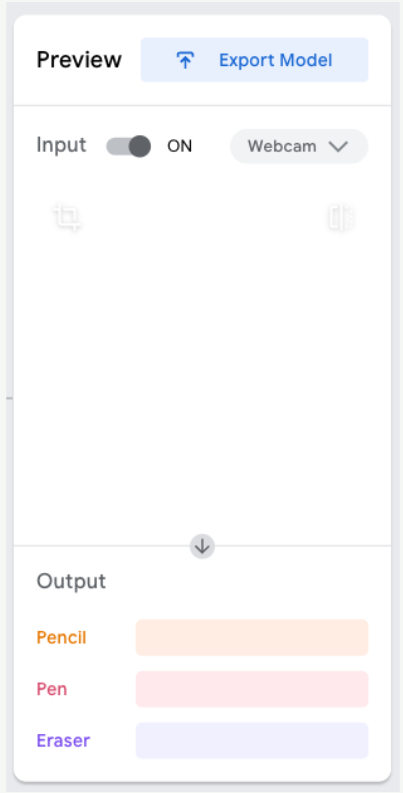
Wait for completion; keep the tab open and the iPad awake

步驟 9：開啟即時相機預覽

Step 9: open the live camera preview

在 Preview 選 Webcam，Input 設為 ON

In Preview, choose Webcam and turn Input ON



允許相機；先用其他組別的物件試取景

Allow the camera; practise framing with another group’s item

若即時相機未能使用

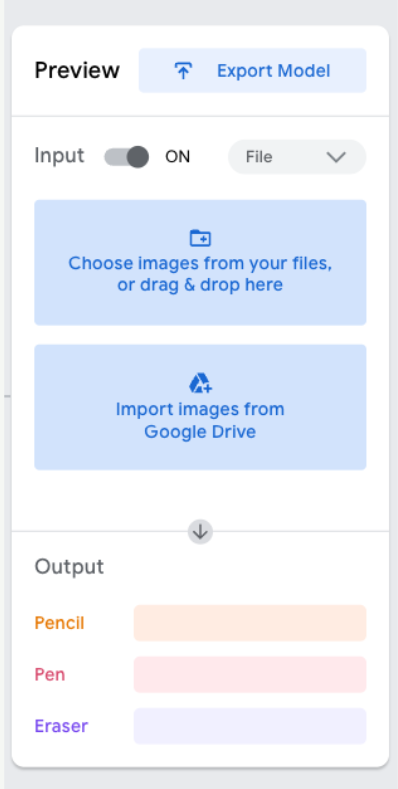
If the live camera is unavailable

請老師檢查相機權限；可改用 Preview 的 File

Ask the teacher to check permission; Preview → File is a backup

只在 Preview 選新的測試照片，不放入訓練

Select a new test photo in Preview only, never in training



照片測試是後備方法，不是即時相機示範

Photo testing is a backup, not a live camera demonstration

步驟 10：投影 iPad 畫面

Step 10: mirror the iPad screen

連接課室 Wi-Fi；由右上角向下滑

Join classroom Wi-Fi; swipe down from the top-right

按螢幕鏡像輸出，選老師指定的接收裝置

Tap Screen Mirroring; select the teacher’s receiver

輸入畫面上的密碼，再回到 Preview

Enter any displayed passcode, then return to Preview

步驟 11：測試自己組別的文具

Step 11: test your group’s unseen stationery

逐一展示自己的鉛筆、原子筆、擦膠

Show your own pencil, pen and eraser, one at a time

每件試上方、正面、側面；停一停讀最高分標籤

Try above, front and side; pause and read the top-scoring label

投影展示；記錄全部結果，包括錯誤

Project the preview; record every result, including mistakes

步驟 12：記下首次測試結果

Step 12: record the first test

每格寫模型預測的類別，答對的加剔號

Write the predicted class in each cell; tick correct results

我們的物件 Our item	上方 Above	正面 Front	側面 Side
鉛筆 / Pencil	_____	_____	_____
原子筆 / Pen	_____	_____	_____
擦膠 / Eraser	_____	_____	_____

答對次數：_____ / 9 個角度；只測了 3 件物件

Correct: _____ / 9 views; only three physical objects were tested

最高分，也可能答錯

A high score can still be wrong

輸出條顯示模型分數，最高分是它的選擇

The output bars show scores; the highest is its choice

高分不保證正確，也不是整體準確率

A high score is not a guarantee or overall accuracy

試空桌或間尺：它仍可能選三類之一

Try an empty desk or ruler; it may still choose one of the three classes

改善圖片，再訓練及比較

Improve the data, retrain and compare

找出一個常見錯誤，保留首次結果

Find one recurring mistake; keep the first results

補拍其他組別文具的角度、光線及背景

Add other groups' items with more angles, lighting and backgrounds

重測作比較；公平新測試要用另一套未見文具

Retest to compare; use a fresh unseen set for a new fair test

小組一分鐘示範

Your group’s one-minute demonstration

說明用了哪些組別的文具訓練

Say which groups’ items you used for training

投影並即時展示自己的三件文具

Project and recognise your own three items live

報告首次測試結果，以及一個錯誤或改善

Report the first results and one mistake or improvement

離堂小結

Exit ticket

回歸和分類有甚麼分別？

How are regression and classification different?

為甚麼同一物件要拍多個角度？

Why photograph the same object from several angles?

為甚麼自己的文具不能用來訓練？

Why keep your own stationery out of training?

教師準備：課前實機檢查

Teacher preparation: check the classroom devices

在學校 iPad 試開網頁、畫數字、上載及訓練

On a school iPad, try both sites, drawing, upload and training

檢查相機預測及課室鏡像投影

Check live camera predictions and classroom mirroring

已檢查桌面瀏覽器；未完成實體 iPad 驗收

Desktop browser checked; physical iPad acceptance is still required

教師流程與器材

Teacher timing and materials

理論及示範約 18 分鐘；收集及訓練約 30 分鐘

About 18 minutes for theory/demo; 30 minutes for data and training

每組示範約 1 分鐘，另預留連接時間

About one minute per group, plus time to connect

iPad、Wi-Fi、投影接收裝置、文具、雙面工作紙

iPads, Wi-Fi, projection receiver, stationery and the two-sided worksheet

資料及圖片來源

Sources and image credits

MNIST：LeCun 等；CNN 歷史：LeCun 等、AlexNet 論文

MNIST: LeCun et al.; CNN history: LeCun et al. and the AlexNet paper

網頁截圖：Teachable Machine、Adam Harley CNN 示範

Live screenshots: Teachable Machine and Adam Harley’s CNN demo

MNIST 圖片：Josef Steppan（CC BY-SA 4.0）

MNIST image: Josef Steppan, CC BY-SA 4.0

截圖日期：2026 年 10 月 12 日；完整來源連結見講者備註

Screenshots: 12 October 2026; full source links in speaker notes